

PREM SAI MUNAGALA

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Profile

I Am An Full Stack Python Developer and Security Analyst with hands-on experience building scalable, secure web applications using Python, NumPy, and Pandas. Proven ability to design end-to-end systems, execute comprehensive ethical hacking and penetration testing strategies. Strong track record of identifying system vulnerabilities, optimizing performance, and turning business requirements into high-performing, resilient technical solutions that improve efficiency, usability, and overall system security.

Education

Parul Institute Of Engineering & Technology

July 2023 – June 2027

Bachelor Of Engineering In Computer Science And Engineering

Gujarat, Vadodara

Experience

Parul Institute Of Engineering And Technology

January 2026 – March 2026

Ethical Hacking & Penetration Testing Intern

Vadodara, Gujarat

- Performed comprehensive penetration testing on full-stack web applications to proactively discover and remediate OWASP Top 10 vulnerabilities. Orchestrated threat modeling and architectural security assessments to enforce robust protection across distributed system components.
- Developed full-stack web applications using Python, Numpy, Pandas, HTML, CSS, JavaScript, and Oracle, implementing the MVC architecture to ensure clean, maintainable, and scalable code.
- Designed and consumed RESTful APIs, integrating them with front-end components.

Projects

Email Security Gateway With Threat Intelligence And Digital Forensics

To overcome the limitations of traditional rule-based spam filters against zero-day phishing and Business Email Compromise, I architected an Enterprise Email Security Gateway featuring a custom Chrome extension for endpoint data acquisition from webmail interfaces. This frontend securely transmits raw message metrics via API to a multi-layered Python and Flask backend that parses complex headers to validate SPF, DKIM, and DMARC protocols. To transition from static defense dependencies to dynamic intelligence tracking, the pipeline isolates and extracts Indicators of Compromise—such as domains, URLs, and attachment hashes—cross-referencing them against global threat feeds via the VirusTotal V3 API. These parameters are evaluated by a custom modular Risk-Scoring Algorithm that aggregates keyword weights, structural URL flaws, and threat intelligence to automate precise 'Safe, Suspicious, or Malicious' verdicts. Furthermore, I implemented a 'Forensics-by-Design' framework using a decoupled, asynchronous SQLite repository to cryptographically hash and log all incidents, establishing an immutable Chain of Custody for audit-ready reconstruction. By decoupling these resource-intensive lookups from the real-time mail stream, this optimization eliminates processing bottlenecks to maximize overall system efficiency.

Technical Skills

- Languages : Python, C, HTML/CSS, Java Script, Boot Strap, Oracle (SQL)
- Developer Tools : Vs Code, PyCharm, Git, Git Hub
- Technologies / FrameWorks : Django, Flask, FastAPI, Numpy, Pandas, Matplotlib

